

A Simulation Toolkit for Synergy-Driven Origami Dexterous Hands

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1. Introduction Replicating human hand dexterity in robotics is a formidable challenge. Fully actuating each joint leads to complex, fragile designs. *Soft synergy* principles [1] enable many-DoF hands controlled by few actuators. Origami fabrication offers a lightweight, low-cost route to realize such synergy-driven hands. However, designing and simulating both the kinematics and contact mechanics of origami structures with tendon transmissions demands a dedicated framework. This project develops an integrated simulation toolkit bridging CAD-level origami hand design with geometry-based synergy solvers and force-based numerical simulation, enabling principled exploration of synergy-based origami hands.

2. Related Work The *adaptive synergy* framework [1] translates postural synergies into hardware: an underactuated hand with a single motor grasping widely via differential transmission and elastic joints. The *augmented adaptive synergy* [2] adds friction-driven actuation (SoftHand 2, in-hand manipulation). The *dynamic synergy* [3] uses passive dampers for speed-dependent power/precision grasps. This work unifies these models in an open-source simulation framework for origami hands, adding force-based grasp simulation beyond existing geometric solvers.

3. Method The toolkit implements four modules:

3.1 Design Parsing & Kinematics. A DXF/JSON parser reconstructs hand topology—fold lines, faces, joints, tendon routings. Kinematics handles tree/cycle-containing origami via BFS spanning trees and iterative relaxation.

3.2 Geometry-Based Synergy Solvers. Three quasi-static models:

- **Adaptive** [1]: $q = E^{-1}R^T(RE^{-1}R^T)^{-1}\sigma$.
- **Augmented** [2]: $\begin{bmatrix} R \\ R_f \end{bmatrix}_E^+ \begin{bmatrix} \sigma \\ \sigma_f \end{bmatrix}$ via Capstan friction.
- **Dynamic** [3]: $E_{\text{eff}}(\alpha) = E + \alpha \text{diag}(T^T C T)$ interpolates power $\leftrightarrow$ precision grasps.

3.3 Force-Based Numerical Simulation. Extends beyond kinematic solvers to full dynamics:

- **Contact-aware equilibrium.** Couples the quasi-static solvers with MuJoCo's physics engine ($M(q)\ddot{q} + h(q, \dot{q}) = \tau + J^T f_c$). Enabling gravity and LCP-based contact resolution produces accurate grasp force distribution. The hand's passive compliance C (from $q = C J^T f_c$ in [1]) is tested against dynamic contact forces.
- **Tendon friction dynamics.** Time-domain integration of the Capstan friction ODE (Eq. 38–42 in [2]) captures sliding transients $\dot{s} \neq 0$ during in-hand manipulation, validating augmented synergy beyond steady-state.
- **Grasp quality metrics.** Computes internal-force subspace Q_f from the grasp matrix ($\delta f_c = Q_f \delta q + Q_u \delta u$ per [1])

and force-closure tests, allowing quantitative comparison of different synergy/routing designs.

3.4 URDF Export & Optimization. Exports URDF+STL for Pinocchio (kinematic) and MuJoCo (physics), featuring interactive synergy sliders. Design optimization searches pulley radii, spring stiffness, damper coefficients via differential evolution to achieve desired synergy directions.

4. Expected Results & Demo Expected outcomes: (i) CAD-to-simulation origami hand design; (ii) validation of simulated synergy directions vs. theory; (iii) quantitative grasp quality evaluation; (iv) optimization pipeline.

Live Demo:

1. Load a 5-finger origami hand into the CAD viewer.
2. Show in MuJoCo with synergy controller: σ (coordinated closure), σ_f (friction differential), dynamic speed slider.
3. Force-based mode: enable gravity, grasp objects, compute contact forces and force closure.
4. (Optional) Run optimization for improved routing.

5. Conclusion This project delivers an open-source toolkit integrating origami hand design, synergy-based underactuation, and both geometric and force-based simulation, accelerating research in low-cost dexterous robotic manipulation.

References

References

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